

A frozen whole-brain *Drosophila* connectome in a closed sensorimotor loop with *Doom*

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Abstract

The synapse-resolution *Drosophila* connectome makes it possible to simulate an entire brain with essentially no free parameters. Whole-brain models of this kind are at present driven *open loop*: sensory neurons are stimulated and motor neurons are read. We embed a frozen FlyWire FAFB v783 connectome (139,255 neurons; 2,710,038 edges at a five-synapse threshold) in a *closed* sensorimotor loop with *Doom*, running at $1.01\times$ realtime on one consumer GPU. Nothing is trained; the only fitted quantity is a single global synaptic gain, calibrated on a non-visual behaviour.

We find a sharp dissociation between modalities. Chemosensory transformations transfer: sugar stimulation drives the proboscis motor pool to 77 Hz, bitter silences it, and co-stimulation is 99% suppressed — none of which was fitted. Adding an olfactory channel raises lateral horn activity from 0.32 to 14.1 ± 5.5 Hz in 10/10 seeds ($p = 0.002$, two-sided sign test) and changes descending-neuron output one synapse downstream, with the sign the wiring predicts (LH $\rightarrow$ DNp01, -248 synapses, suppression; -17.7 ± 7.4 Hz, 9/10 seeds, $p = 0.021$), while a degree-preserving shuffle of the same graph produces no such effect.

Visual motion computation does not transfer. Direction selectivity and looming discrimination both fail, and we show the failure is a property of the model class rather than of the connectome: under linear synaptic summation with a single output threshold, $\text{thr}(A + B)$ is symmetric in A and B , so temporal order is not recoverable. The correlator geometry is verifiably present in the graph and a retinotopic travelling wave reaches **T4a**, yet six candidate remedies — per-type conduction delays, heterogeneous membrane time constants, conductance-based shunting, a forced high-conductance state, population pooling, and graded non-spiking optic-lobe units — yield at best $|\text{DSI}| = 0.0121$, some $40\times$ below the threshold conventionally used to *select* direction-selective terminals experimentally, with a mirror-pair sign relation no better than chance across the operating range. The residual obstacle is measurable: **T4a**’s fast excitatory arm is starved 37:1 by glutamatergic L1 inhibition.

Read against connectome-constrained models whose unknown parameters are optimized, and which do recover motion selectivity, this localizes what that optimization contributes: the wiring is common to both, so what optimization supplies is the per-type gain and timing structure that balances the correlator’s arms. We additionally report a methodological result — a degree-preserving shuffle control is invalid in closed loop, because the two arms generate different stimulus distributions.

1 Introduction

An electron-microscopy reconstruction of an adult *Drosophila* brain now provides every chemical synapse between all 139,255 neurons [Dorkenwald et al., 2024], together with cell-type annotation

[Schlegel et al., 2024] and per-synapse neurotransmitter prediction [Eckstein et al., 2024]. This raises a question that was previously unaskable: *how much of an animal’s behaviour is implied by its wiring alone?*

Two recent lines of work answer parts of it, and they answer differently. Shiu et al. [2024] built a whole-brain leaky integrate-and-fire model with a uniform parameter set and a single fitted scalar — the per-synapse weight W_{syn} — and recovered gustatory behaviours such as proboscis extension. Lappalainen et al. [2024] built a connectome-*constrained* model of 64 optic-lobe cell types in which the unknown biophysical parameters were *optimized* by gradient descent on a motion-estimation task, and recovered single-neuron responses matching 26 experimental studies. These differ in two ways at once: whole-brain versus optic lobe, and frozen versus optimized.

Both are driven open loop. Wang-Chen et al. [2024] close the loop, embedding the optimized visual network of Lappalainen et al. [2024] in a biomechanical fly with reinforcement-learned controllers. The configuration that has not been tested is the remaining corner: a *whole-brain, frozen, unoptimized* connectome placed in a closed sensorimotor loop.

That corner is informative precisely because of the contrast. If a frozen whole-brain model reproduces a computation, the wiring plus a global gain sufficed. If it fails at a computation that the optimized model achieves on the same wiring, the difference isolates what optimization supplied.

We report such a test. We drive the frozen connectome with the rendered frames of *Doom* [Kempka et al., 2016], read eight characterized descending neurons, and return their activity to the game as control input, in real time. The environment is adversarial and non-stationary but — crucially for interpretation — entirely outside the fly’s evolutionary experience, so nothing here is a test of ecological competence. It is a test of which sensorimotor transformations survive the transfer from wiring diagram to running system.

Contributions.

1. A real-time closed-loop embodiment of a frozen whole-brain connectome ($1.01\times$ realtime, 139,255 neurons, single consumer GPU), released as open code.
2. A modality dissociation: chemosensory transformations transfer, visual motion computations do not.
3. A mechanistic account of the visual failure, with six candidate remedies implemented and measured, isolating a 37:1 imbalance in the elementary motion detector’s two arms.
4. A positive result that beats a degree-preserving shuffle: an olfactory channel reaches descending neurons through the anatomically named route with the wiring-predicted sign.
5. A methodological caution: shuffle controls are invalid in closed loop.

2 Results

2.1 A frozen whole-brain connectome running in closed loop

We load FlyWire FAFB v783 and retain edges of at least five synapses, giving 2,710,038 edges carrying 31,578,726 synapses among 134,013 connected neurons (Table 1). Signs follow the predicted transmitter, with glutamate treated as *inhibitory* — it acts through GluCl chloride channels in flies [Cleland, 1996] and accounts for 16.6% of the edges we simulate (19.5% before thresholding), so importing the vertebrate convention would mis-sign about a sixth of the brain.

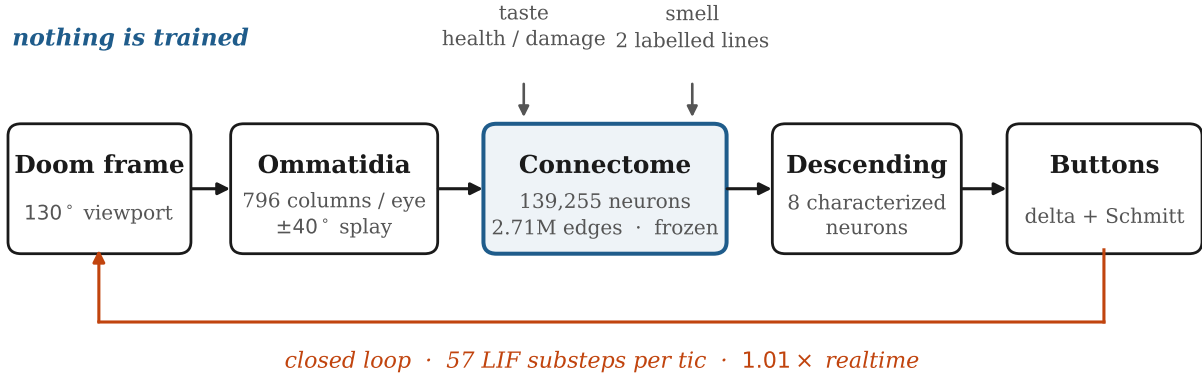


Figure 1: The closed loop. Rendered frames are sampled by the real ommatidial lattice, propagated through the frozen connectome for 57 leaky integrate-and-fire substeps per game tic, and read out at eight characterized descending neurons whose activity becomes the next action. Taste and smell enter as direct drive on gustatory and olfactory receptor neurons. No component is trained and no parameter is fitted to the environment.

We sign per edge rather than per neuron, and in this pipeline the choice is not free. Transmitter is predicted per synapse and reported per neuropil, so one neuron can carry different predictions in different neuropils: 42,078 of the 128,972 presynaptic neurons in the simulated graph (32.6%) carry more than one, and 26,781 (20.8%) carry both signs. Collapsing each neuron to a synapse-weighted majority — the usual way of imposing Dale’s law — would flip 3.1% of edges and 2.6% of synapses. This matters for reading the literature as much as for the model: the alternative Princeton release has Dale’s law applied upstream (0 of its 139,003 presynaptic neurons carry more than one transmitter), so per-neuron and per-edge signing coincide there and not here, and transmitter statistics quoted from one release do not describe the other.

Photoreceptors required an explicit override. They release histamine [Hardie, 1989], which is not among the six transmitters the classifier predicts; on R1-6 output edges the classifier returns ACh 71.2%, Glu 20.2%, GABA 5.8%, and every excitatory call yields the wrong sign for a histaminergic chloride synapse. We force all photoreceptor output inhibitory, flipping 11,920 of 18,437 edges. Without this the optic lobe processes a contrast-inverted image.

Neurons are leaky integrate-and-fire with the parameters of Shiu et al. [2024], themselves drawn from Kakaria and de Bivort [2017]: $V_{\text{rest}} = -52$ mV, $V_{\text{thresh}} = -45$ mV, $\tau_{\text{mem}} = 20$ ms, $\tau_{\text{syn}} = 5$ ms, $t_{\text{refrac}} = 2.2$ ms, $t_{\text{delay}} = 1.8$ ms. We integrate the linear subthreshold system exactly rather than by forward Euler, which we measured 6% low on a single postsynaptic potential. At $dt = 0.5$ ms this gives 57 substeps per game tic, with the frame held across all 57.

Vision enters through the real retinotopy: `column_assignment` supplies 796 ommatidial columns per eye on a hexagonal lattice. Each column samples the rendered frame through a Gaussian acceptance function, implemented as a separable pre-blur followed by bilinear sampling at that column’s gaze direction. The engine draws a planar perspective projection, so a column at azimuth θ is sampled at screen coordinate $\tan \theta / \tan(\text{FOV}/2)$ rather than linearly in θ ; the linear map misplaces a column at 10° of gaze by 87% and one at 30° by 71%, converging only at the very edge of the viewport. Since retinotopy is the substrate every motion computation runs on, and a warp of that size makes the angular spacing between neighbouring columns vary about twofold across the field, we

Table 1: Graph as simulated. Unthresholded synapse count matches the figure published for this release, confirming the extraction.

Neurons (annotated)	139,255
Edges, unthresholded	16,847,997
Synapses, unthresholded	54,492,922
Edges simulated (≥ 5 synapses)	2,710,038
Synapses simulated	31,578,726
Neurons with ≥ 1 edge	134,013
Graded (non-spiking) units	77,873
Resident GPU memory	31 MB
Wall-clock per 0.5 ms step	$\sim 300 \mu\text{s}$
Closed-loop rate	$1.01 \times \text{realtime}$

note it as a correctness requirement rather than a refinement: rigid motion of the world then sweeps the retina at a speed that depends on where one looks, which is precisely the input a correlator tuned to a fixed spacing cannot use. The two eyes are splayed $\pm 40^\circ$; collapsing them onto the screen centre makes both eyes see the same image and drives the left-right steering difference to zero by construction. The engine’s field of view must be set through the console rather than the command line — passing `+fov` is silently ignored, and renders at 90° , 130° and 160° are bit-identical — and it resets with each episode. At 130° the viewport covers 64% of the lattice and at 160° , 80%; remaining columns are filled with the frame’s mean luminance, because a dark surround is a permanent high-contrast edge that the looming detectors report as a stationary object.

Frames are not held across the substeps of a tic. Doom draws at 35 Hz and the simulator runs 57 substeps per frame, so a held frame presents the retina with a step change followed by 56 substeps of exactly zero temporal derivative, while the delayed arm of the correlator is 80 ms, or three frames — that is, an impulse train aliased against the detector’s own delay line. Because a fly in a room receives continuous motion and the strobe is an artifact of the game engine rather than of the biology, we interpolate luminance linearly between consecutive frames across the substeps. This also makes the retina’s Weber adaptation advance once per substep as its time constant assumes; stepping it once per tic instead gave it an effective τ of 14 s in place of 0.25 s.

Output is read from eight characterized descending neurons — DNa02 for steering [Rayshubskiy et al., 2025], BPN for forward walking [Bidaye et al., 2020], MDN for backward walking [Bidaye et al., 2014], DNp01 (the giant fibre) for escape [von Reyn et al., 2014] — plus the proboscis motor pool and gustatory receptor neurons. The graded channels use the engine’s delta buttons; binary actions use a two-threshold Schmitt trigger.

Two of the graded channels read a *difference between the two sides* of a bilaterally paired cell type, and there the standing asymmetry of a single reconstructed brain is a confound rather than a command: DNa02 sits at 150 against 257 Hz and DNp01 at 110 against 206 Hz, both of which exceed their command clamps and pin the button. We therefore subtract a slow baseline ($\tau = 3$ s) from the paired channels before decoding, keeping the transients and discarding the DC, and set each gain so that three standard deviations of what remains reach the clamp. The forward channel is not a bilateral pair — it is BPN – MDN, two opposing cell types — so its standing +27.6 Hz is a tonic walk command and is kept, with the gain placing it at half the clamp instead. We note the consequence rather than manage it: that difference carries ± 1.3 Hz of modulation on a 27.6 Hz mean, so walking speed here is 4.6% modulated and effectively tonic.

Table 2: Looming (M4), open loop, mean rate in Hz. Neither arm separates looming from receding, and the intact connectome does not outperform a degree-preserving shuffle.

	Intact			Shuffled		
	LC4	LPLC2	DNp01	LC4	LPLC2	DNp01
Looming	36.06	0.00	139.0	7.04	2.82	88.7
Static	73.37	0.02	149.3	45.15	14.41	207.0
Receding	34.58	0.00	149.7	6.19	2.50	84.0

2.2 Chemosensory transformation transfers

As a positive control we reproduce the gustatory result of [Shiu et al. \[2024\]](#). Stimulating sugar receptor neurons at 100 Hz drives the proboscis motor pool to 77 Hz; stimulating bitter receptors drives it to zero; stimulating both together yields 99% suppression relative to sugar alone.

This control has a known circularity that we state rather than manage: like [Shiu et al. \[2024\]](#) we fit W_{syn} on this very behaviour, obtaining 0.1655 mV against their 0.275 mV (the difference is attributable to a different connectome release and synapse pipeline). What the test genuinely establishes is therefore weaker than it appears — that *some* single scalar gain reproduces the behaviour — but it is not vacuous: the bitter–sugar suppression ratio was not fitted, and a sign-scrambled graph admits no such scalar.

2.3 Visual motion computation does not transfer

Direction selectivity fails. Presenting drifting gratings and measuring the direction selectivity index $\text{DSI} = (R_{\text{pref}} - R_{\text{null}})/(R_{\text{pref}} + R_{\text{null}})$ in T4/T5, the canonical elementary motion detectors [[Maisak et al., 2013](#)], gives $|\text{DSI}| \leq 0.0121$. Experimental studies routinely *select* terminals for analysis at $\text{DSI} > 0.5$, so the model sits some $40\times$ below the threshold at which a real terminal would merely qualify as direction-tuned.

Looming fails in the same way and more starkly (Table 2). Neither the intact nor a degree-preserving shuffled connectome distinguishes an expanding from a contracting stimulus; both instead show static $\gg$ looming $\approx$ receding, the signature of a response to dark *area* rather than to expansion. Because the shuffled graph performs comparably, the measurable looming response is not attributable to connectivity at all.

The looming detector is not weakly driven; it is held shut. In closed loop LPLC2 reads exactly 0.00 Hz, and the reason is measurable. It sits at -52.2 mV — *below* its own resting potential and 7.2 mV from threshold — because its heaviest input, PVLP011 at -45.6 synapses per cell, fires at 371 Hz. The model’s absolute refractory ceiling is $1/t_{\text{refrac}} = 455$ Hz, so that one inhibitory cell runs at 82% of the maximum rate the model permits, and the inhibitory conductance it delivers exceeds the excitatory arm (Tm5f, $+44.7$ synapses per cell but only 36 Hz) by roughly tenfold. This is the T4a arm imbalance again, in a second and independent circuit.

That it is an imbalance rather than a shortage is what makes it diagnostic. We swept every input-side scale available — photoreceptor drive, the graded rate ceiling, and Weber contrast gain — and LPLC2 does not leave 0.00 Hz at any setting. Reducing drive lowers PVLP011 from 371 to 113 Hz but takes LC4 from 53 to 0.0 Hz with it. The result is expected on inspection: an input scale multiplies both arms of a ratio and cancels, so no common gain can change a 10:1 imbalance. Only a gain that acts differentially on the two arms can, which is to say a per-type gain — the quantity the connectome does not specify and the optimized model of [Lappalainen et al. \[2024\]](#) obtains by fitting.

A tonic depolarising drive to the optic lobe does release it, and shows the same trade rather than escaping it. Raising the bias from 0 to 6 mV lifts LPLC2 from 0.0 to 64.7 Hz, but it also lifts MDN, and the walking command is BPN – MDN: that difference falls from +22 Hz to +0.3 Hz by 2 mV and reverses sign by 6 mV, at which point the model walks backwards. Meanwhile the population above 200 Hz grows from 197 cells to 991. There is no setting at which the looming pathway conducts and locomotion survives.

A blank-field control shows the trade is worse than that, and it is the reason we report bias sweeps with one. Presenting an *empty screen* at 7.5 mV bias drives LPLC2 to 76.17 Hz, against 76.22 Hz for an expanding disc — a stimulus contribution of 0.07%. At 4 mV it is 30.52 blank against 30.62 looming. The bias that releases the cell is the same quantity that determines its output: past threshold we are no longer measuring vision but measuring the bias, and the fraction of the response attributable to the picture *falls* as the bias rises. There are two regimes and neither is looming detection — LPLC2 silent, or LPLC2 firing at a rate the stimulus barely perturbs. Reporting the looming response alone, without the blank, would have made 76 Hz look like success.

We also tested whether the failure is an artefact of where visual drive enters. By default we inject at the lamina monopolars, one synapse past the photoreceptors, which bypasses two real features of the eye: neural superposition, in which six photoreceptors sharing a visual direction converge on one cartridge, and the histaminergic first synapse that inverts the signal. Injecting at R1-6 instead restores both. It changes nothing. At matched bias the two sites agree to within the blank-field offset (LPLC2 18.4 against 30.6 at 4 mV, 76.2 against 76.4 at 7.5 mV), and at zero bias photoreceptor injection is silent throughout, because the inhibition-dominated optic lobe has no baseline to disinhibit from. The one thing photoreceptor injection does carry is a bias of its own: FAFB’s uneven optic-lobe proofreading recovers a column for 451 of 785 left columns against 749 of 796 right, so it is unusable for any measurement of a left–right difference, which is why the default sits at the lamina.

Across every condition in that sweep, looming and receding differ by less than the blank-field offset, and the *sign* of the difference flips between conditions. The only regime with genuine stimulus-driven structure is zero bias at the lamina, where LC4 gives 77.2 Hz for a static disc against 24.9 Hz blank — a real threefold response, to dark *area*. The model reports how much dark is present, never whether it is growing.

Downstream visual functions fail correspondingly. LPLC2 activity is uncorrelated with true angular size ($r = +0.000$), and no azimuth signal exists anywhere we could find: neither LC11 ($r = +0.034$) nor LC4 ($r = +0.070$) carries target bearing. LC10a, the cell type that would carry it, is silent at 0.00 Hz, receiving only +45/ – 83 synapses per cell against LC11’s +477/ – 456 and LC4’s +439/ – 259; it is held down by AOTU042 and TuTuAa. Its known state-dependent gain boost is a *male* courtship mechanism gated by P1, and FAFB is a female brain, so that circuit is absent. Nor is there a state signal to substitute for it. AOTU042 suppresses LC10a at –18 synapses per cell, but pC1 and AOTU042 are not connected in this brain at all — zero edges in either direction, at any synapse threshold — so the aggression state cannot reach that suppression to lift it or to deepen it.

2.4 Why motion fails: temporal order, and a starved arm

Direction selectivity and looming selectivity are both computations over the *temporal order* in which neighbouring retinotopic columns activate. Motion to the right makes column *A* lead column *B*; motion to the left makes *B* lead *A*. The two cases contain identical events and differ only in order.

In the model, presynaptic inputs sum linearly into a conductance and the only nonlinearity is a threshold on that sum. But $\text{thr}(A + B)$ is symmetric in its arguments, so a unit that sums its inputs and thresholds the sum cannot distinguish the two cases. This is not a tuning deficiency;

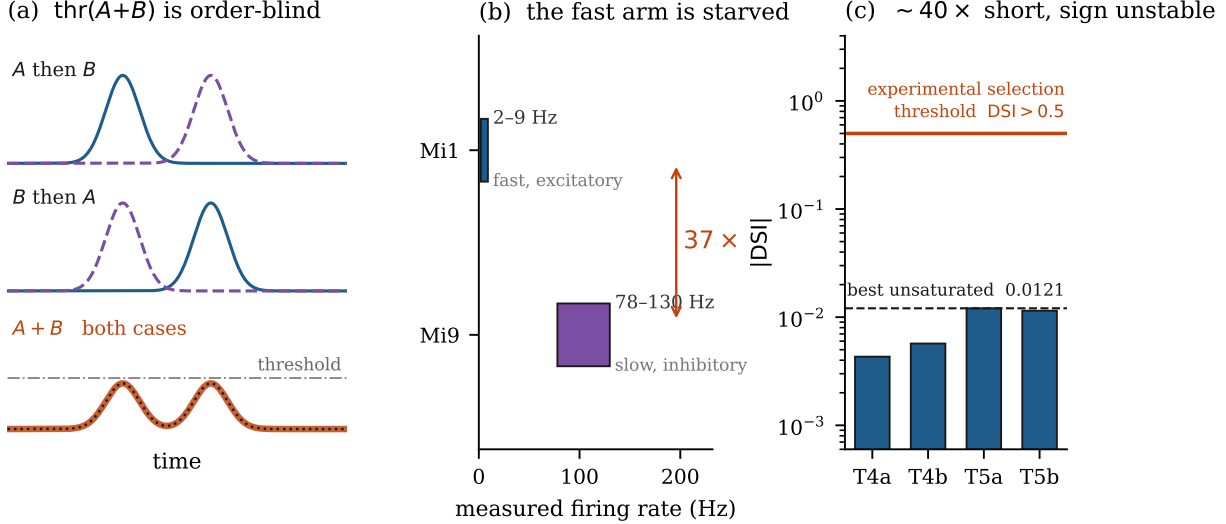


Figure 2: Why motion fails. (a) A unit that sums its inputs and applies one threshold to the sum cannot recover order: “A then B” and “B then A” produce identical sums (rust curve, with the second case overlaid dotted), so any threshold on that sum responds identically. (b) Measured firing rates of T4a’s two correlator arms: the fast excitatory arm (Mi1) is suppressed by glutamatergic L1 while the slow inhibitory arm (Mi9) runs freely, a $37\times$ conductance imbalance. (c) Resulting direction selectivity with graded optic-lobe units. Magnitudes fall some $40\times$ short of the value at which experimental studies merely *select* a terminal as direction-tuned, and across 45 unsaturated operating points the two mirror pairs oppose at only 3, which is below chance — the sign is fluctuating, not small-but-real.

it is a structural property of the model class. Biological elementary motion detectors escape it by delaying one arm and combining the arms *multiplicatively* [Hassenstein and Reichardt, 1956, Borst et al., 2010], realized in *Drosophila* as preferred-direction enhancement together with null-direction suppression [Haag et al., 2016, Gruntman et al., 2018].

Both ingredients are present in the graph, and we verified them. T4a receives fast centred excitation (Mi1 +30.6, Tm3 +7.0) and slow flanking inhibition on opposite sides (Mi9 −7.4 at (−0.76, +0.26); Mi4 −3.4 at (+0.39, −0.60)), with four subtypes on four mirrored axes. The stimulus reaches them: L1’s response phase advances at −20.0 deg/deg against an expected 20.0, and this retinotopic travelling wave survives to T4a.

We implemented and measured six candidate remedies. Five produced nothing: per-type conduction delays (1.8–240 ms), per-neuron membrane time constants (5–200 ms), conductance-based shunting synapses, a forced high-conductance state ($g_{\text{tot}} = 105$, two orders of magnitude past the linear regime), and population pooling (per-cell DSI against per-cell input offset, $r = +0.043$ over 874 cells). The sixth — modelling the optic lobe as *graded, non-spiking* units, which is what photoreceptors and lamina monopolar cells actually are — was the only one to produce a measurable effect at all (Table 3), and it is small enough that its sign has to be checked rather than assumed.

Checking it requires sweeping the operating point rather than picking one, because the dominant influence on the number is not the wiring: graded units are capped at 200 Hz, and a unit against its ceiling reports $\text{DSI} \approx 0$ whatever its inputs do. We therefore swept bias, spatial period and temporal frequency (72 points), kept the 45 below 75% saturation, and report the largest effect among them. Two things follow. The magnitude is $|\text{DSI}| \leq 0.0121$, some $40\times$ below the value at which experimental studies merely *select* a terminal as direction-tuned. And the qualitative signature is absent: both mirror pairs oppose at only 3 of the 45 unsaturated points, where two independent

Table 3: Direction selectivity, at the strongest operating point that is not saturated. Graded units are capped at 200 Hz and a unit against its ceiling reports $DSI \approx 0$ whatever its inputs do, so we sweep bias, spatial period and temporal frequency (72 points), keep the 45 below 75% saturation, and report the largest effect among them — bias 1.0 mV, period 30° , 1 Hz. The magnitude is $41\times$ below the value at which experimental studies merely *select* a terminal as direction-tuned. The sign is not reliable either: both mirror pairs oppose at only 3 of 45 unsaturated points (7%), by pair T4a/T4b 5/45, T5a/T5b 15/45. A correlator’s sign should not depend on the operating point, so this is a quantity fluctuating about zero rather than a small but real selectivity.

Pathway		DSI		Mirror pair
ON	T4a	−0.00431	T4b −0.00570	same sign
OFF	T5a	−0.01208	T5b −0.01147	same sign

coin flips would give about 11. A correlator’s sign does not depend on its operating point, so what we are measuring is a quantity fluctuating about zero rather than a small but real selectivity. An earlier version of this manuscript reported opposed mirror pairs from a single configuration; the sweep does not support that claim and we withdraw it.

The residual obstacle is measurable and specific: the fast arm is starved. Weighting T4a’s inputs by measured firing rate rather than synapse count gives excitation 189 against inhibition 558, a $3.0\times$ imbalance in drive; the realised synaptic conductances are $g_e = 0.0018$ against $g_i = 0.0667$, a ratio of $37\times$. The two quantities measure different things — rate-weighted input against the conductance it actually produces — and it is the second that sets how far the excitatory arm can move the membrane. Mi1 sits at 2–9 Hz because L1 suppresses it (−77.3, glutamatergic and therefore inhibitory), while Mi9 runs at 78–130 Hz. The ON pathway is disinhibitory and, with uniform per-type gains, never rebalances. The OFF pathway, which does not depend on that disinhibition, conducts normally (T5a 58 Hz against T4a 2 Hz) — an internal control indicating the deficit is pathway-specific rather than a global failure of the optic lobe.

2.5 Olfactory valence transfers, and beats a shuffled control

The environment renders light and nothing else, but a fly in a room with a large animal in it smells that animal; simulating vision alone makes the model impoverished rather than conservative. We therefore added a channel feeding two real labelled lines — ORN_DA1, carrying the pheromone cVA [Kurtovic et al., 2007], and ORN_DM1, carrying vinegar [Semmelhack and Wang, 2009] — with enemies mapped to the former and health pickups to the latter.

The channel is deliberately built weak so that it cannot substitute for vision. Most importantly it carries *no direction*: a fly’s antennae are ~ 0.3 mm apart and cannot triangulate, so both sides receive bit-identical drive. It is additionally saturating, slow ($\tau = 0.2$ s), patchy (3 Hz bursts at 60% duty), and persistent ($\tau = 1.5$ s), so a source moving out of sight still smells.

Testing this in closed loop would be invalid (Section 2.6), so we ran it *open* loop: the agent is held still and the game stepped with a null action, so enemies still move and the scene still evolves, but it evolves identically in both arms because the seed is fixed and nothing feeds back. Smell is then the only difference between arms, and LC4 — a visual cell that should not move — serves as a built-in check that the comparison is sound.

The lateral horn — the innate-valence pathway [Jefferis et al., 2007, Dolan et al., 2019] — is dormant at 0.32 Hz without smell and rises to 14.1 ± 5.5 Hz with it, in all ten seeds (Table 4). The effect reaches the descending neurons in *one synapse*, and its sign is predicted by the wiring: $LH \rightarrow DNp01$ is −248 synapses, hence inhibitory, and $DNp01$ is suppressed by 17.7 ± 7.4 Hz, in nine

Table 4: Effect of the olfactory channel, open loop, 10 seeds $\times$ 320 tics. Δ is the mean on-minus-off difference in Hz; “sign” counts seeds agreeing with the mean direction, with p from a two-sided sign test. In the intact connectome every motor and valence population moves consistently; under a degree-preserving shuffle of the same graph every effect falls to the 0.00–0.11 Hz scale of the LC4 control. Sign consistency alone does not separate the arms — the shuffle retains some of it — so the comparison to read is magnitude. LC4 is a visual cell whose activity should be unaffected by smell; it carries a small offset in both arms, at 10479 $\times$ smaller magnitude than the lateral horn response it bounds.

Population	Intact				Degree-preserving shuffle			Role
	Δ (Hz)	s.d.	sign	p	Δ (Hz)	s.d.	sign	
LHN	+14.06	5.46	10/10	0.002	+0.01	0.01	8/10	innate valence
DNp01	−17.67	7.42	9/10	0.021	−0.00	0.27	6/10	escape (giant fibre)
DNa02	+11.68	5.29	9/10	0.021	+0.11	0.34	7/10	steering
BPN	+1.21	0.46	10/10	0.002	+0.07	0.04	10/10	forward walking
MDN	−0.96	0.40	10/10	0.002	−0.02	0.15	6/10	backward walking
LC4	−0.00	0.02	8/10	0.109	+0.07	0.03	10/10	<i>visual control</i>

seeds of ten. Meanwhile LC4, a visual population that should not move at all, does not: it shifts by -0.00 ± 0.02 Hz, four orders of magnitude below the lateral horn response it bounds.

Crucially, a degree-preserving shuffle of the same graph, run through the identical protocol and the identical seeds, does not reproduce this (Table 4, right). The valence and steering effects collapse by two to three orders of magnitude — LHN from +14.06 to +0.01 Hz (1563 $\times$), DNa02 102 $\times$, DNp01 to within noise of zero — landing at the 0.00–0.11 Hz scale of the LC4 control, that is, at the network’s generic coupling floor.

We want to be precise about what does and does not discriminate here. Sign consistency does *not*: several shuffled populations remain fairly sign-consistent across seeds, including LC4 at 10/10 — the one population that should not be responding at all. What discriminates is magnitude. A sign test on a 0.01 Hz effect is a statement about a systematic that is three orders of magnitude too small to matter, and we read it that way. We also note that the shuffled DNp01 residual reversed sign between two independent executions of the protocol, where every intact effect held its direction — the behaviour expected of noise rather than of a transformation. The honest summary is that the intact effects are large and reproducible while the shuffled ones sit at the control’s floor. *The effect requires the wiring.*

The BPN and MDN walking effects deserve a weaker claim than the rest. They are consistent in direction across all ten seeds, but at ± 1 Hz they are only $\sim 17\times$ above the shuffled floor rather than $\sim 100\times$, and we would not rest anything on them alone.

The shuffle also changes the baseline in an informative way. Without smell the intact lateral horn sits at 0.32 Hz; the shuffled one idles at 25.7–25.8 Hz across seeds, an 80 $\times$ higher floor. Preserving degree while destroying specificity does not merely remove the response — it removes the *quiet*. The intact lateral horn is selectively silent and selectively driven, which is the behaviour a labelled-line valence pathway should show, and neither property survives rewiring.

Two caveats attach permanently. Odour source distances are read from the engine’s label buffer, so the model was *told* that enemies exist; this is evidence about what the connectome does with a signal, not evidence that it can detect one. And 1,227 lateral horn neurons pooled together mixes many functional channels, so “the lateral horn responds” is coarser than it sounds.

A path that exists is not a path that carries. We first routed the same odour at the pC1 aggression neurons, on the strength of a three-hop path touching 8 of 10 cells, and it failed flat

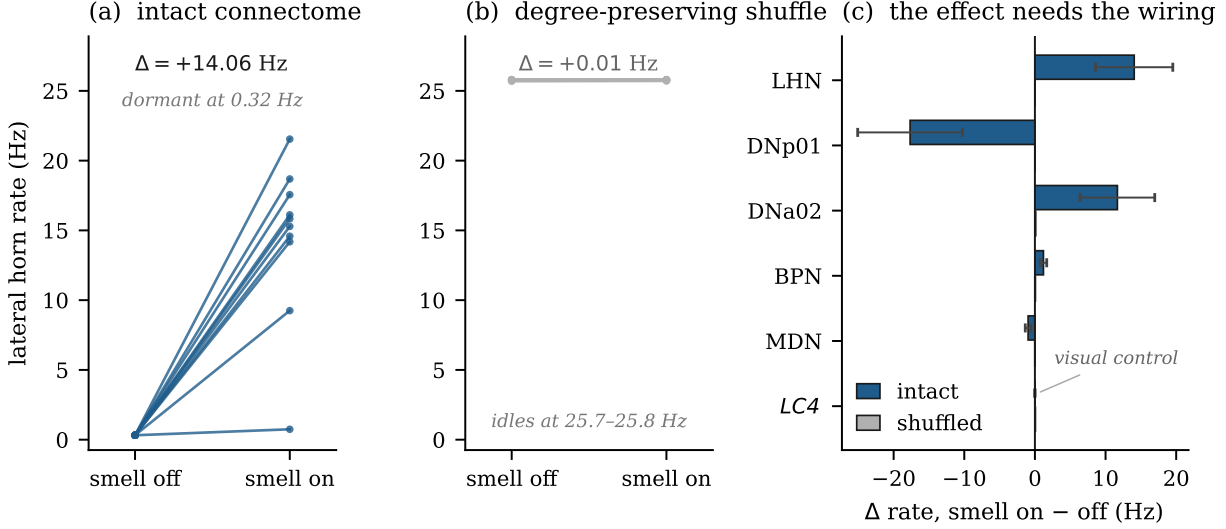


Figure 3: Olfactory valence reaches the descending neurons, and needs the wiring to do it. **(a)** Per-seed lateral horn rate with smell off and on in the intact connectome; all ten seeds rise. **(b)** The same protocol on a degree-preserving shuffle of the same graph: no seed moves appreciably, and the shuffled baseline idles near 21 Hz where the intact pathway is dormant at 0.32 Hz. **(c)** Mean change per population, both arms (error bars are s.d. over ten seeds). LC4 is a visual cell included as a control: it should not move, and does not.

— 0.00 Hz across a tenfold sweep of drive. Measured afterwards, the reason is structural: DA1’s contribution is a rounding error against pC1’s balanced +530/−555 of input, and pC1 is a bistable latch that must be pushed over rather than nudged. The lateral horn is the opposite on every count — it receives 21% of the odour relay’s entire output, has no latch, and sits one hop from the output. Hop count is a poor proxy for influence in a connectome; synaptic weight against competing input is the quantity that matters.

2.6 Shuffle controls are invalid in closed loop

Degree-preserving shuffles are the standard control for attributing an effect to connectivity, and we use one above. But in a closed loop they silently stop being a control.

We measured this directly. Over a matched episode the intact agent encountered an enemy on 186 of 500 tics and finished at full health; the shuffled agent encountered one on 489 of 492 tics and finished at 16 health. The two arms behaved differently, and because behaviour determines what the agent sees, they were no longer sampling the same stimulus distribution. Any cross-arm correlation then compares two different worlds rather than two different brains.

The consequence generalizes to any embodied connectome study: *controls must hold the stimulus fixed*, which in practice means running the comparison open loop, as we do for the olfaction result.

3 Discussion

The dissociation we observe is not between easy and hard computations, nor between small and large circuits. It is between computations that a linear summation followed by a threshold can express and those it cannot. Gustatory and innate-olfactory valence are, at the level the connectome constrains them, weighted sums with signs: sugar excites a motor pool, bitter inhibits it, an odour drives a

valence region that inhibits an escape neuron. Those transfer. Motion direction is a product of a delayed and an undelayed signal, and it does not.

This is worth stating carefully, because it is easy to over-read as a claim about connectomics. The connectome contains the correlator: we found the geometry, the mirrored axes, and the travelling wave that drives them. What it does not contain, and cannot, is the per-type gain and timing structure that makes the two arms comparable in magnitude. In our hands the arms differ by $37\times$, and the $\sim 40\times$ subthreshold, sign-unstable residue that survives is what a correlator produces when one of its inputs is whispering.

Placed beside Lappalainen et al. [2024], this becomes informative rather than merely negative. That model shares our wiring and recovers motion selectivity, but obtains its unknown parameters by optimization. Ours fixes them to a single published set. The difference between the two outcomes is therefore a measurement of what the optimization supplies — and our arm-imbalance analysis suggests a concrete hypothesis about what that is: relative gain along the ON pathway’s disinhibitory chain. That hypothesis is testable by inspecting the learned parameters of the optimized model, which we have not done and which would be the natural next step.

Two further readings we want to foreclose. First, none of this is a claim about whether a fly brain can play a video game; the environment is outside the animal’s evolutionary experience and the model has no ventral nerve cord, so behavioural competence was never the measurable. Second, the olfaction result is not evidence of enemy detection — the odour magnitude is supplied by the engine. What both results measure is transfer: whether a transformation implied by the wiring appears in a running system.

Limitations

- **No ventral nerve cord.** FAFB is brain-only, so the game’s movement code substitutes for legs. All behavioural *timescales* are the engine’s; claims are restricted to descending-neuron activity.
- **No gap junctions.** EM reconstruction captures chemical synapses; the giant fibre’s fastest output is electrical and therefore invisible here.
- **No neuromodulation.** Dopamine, octopamine and serotonin are flattened to fast excitation.
- **$n = 1$ brain, and its halves are not mirror images.** DNa02 sits $\sim 2\times$ higher on the right than the left, which is a standing turn command until removed by a slow baseline.
- **Angular scale is calibrated, not measured.** The lattice is real but its axes are not isotropic in visual angle, so each is scaled to the published field of view. Absolute angular claims inherit this.
- **The gustatory control is partly true by construction,** since W_{syn} is fitted on it.
- **Single stimulus environment.** All visual results come from one rendering engine with one viewport geometry; we have not tested whether a different visual statistic changes the arm imbalance.

4 Methods

Data. FlyWire FAFB v783 [Dorkenwald et al., 2024, Schlegel et al., 2024], frozen October 2023. Connectivity from the Buhmann synapse pipeline [Buhmann et al., 2021] with transmitter predictions from Eckstein et al. [2024]. Edges thresholded at five synapses. Cell types from the consolidated

typing table; BPN and aIPg exist only in the free-text community label file. The data are CC BY-NC 4.0 and are not redistributed with our code.

Neuron model. Leaky integrate-and-fire with parameters as listed in Section 2.1. The subthreshold system $\tau \dot{v} = (V_{\text{rest}} - v) + g$, with g decaying at τ_{syn} , is integrated exactly:

$$v(t+dt) = v_0 A + g_0 B + g_{\text{ext}}(1 - A), \quad A = e^{-dt/\tau_{\text{mem}}}, \quad B = \frac{\tau_{\text{syn}}}{\tau_{\text{syn}} - \tau_{\text{mem}}} (e^{-dt/\tau_{\text{syn}}} - A),$$

matching the scheme used by Brian 2 [Stimberg et al., 2019], against which we validate to 0.03%.

Departures from the reference model. Each is switchable and each carries a measured justification. (i) *Histamine override* on photoreceptor output, as above. (ii) *Conductance-based synapses*: $\tau \dot{v} = (V_{\text{rest}} - v) + g_e(E_e - v) + g_i(E_i - v) + I$ with $E_e = 0$, $E_i = -70$ mV, verified divisive (shunting reduces input-output slope by $> 40\%$ where subtraction preserves it). (iii) *Per-type conduction delays* via a multi-tap ring buffer, since a single global delay cannot express a correlator. (iv) *Graded non-spiking units* for all 77,873 optic neurons, output rectified-linear in $(v - V_{\text{rest}})$ capped at 200 Hz. (v) *Weber contrast adaptation* per column. (vi) L3 kept sustained while L1/L2 adapt — from a stationary agent the rendered scene changes by 0.0003 in mean luminance between frames, so a fully adapting retina washes out and the loop cannot start.

The environment, and what we changed about it. Modifying the game is not a modification of the model, and three properties of Doom are wrong for this animal in ways that suppress the signal under test. *Gamma*: the engine writes gamma-encoded sRGB, but photoreceptors integrate linear flux and the acceptance function is a spatial integral over it, so the Gaussian pre-blur must run in linear light; undoing the encoding first raises retinal contrast by 37%. *Spectrum*: R1-6 carries Rh1, which is blue-green sensitive and nearly blind at Doom’s red primary, while Doom is painted in browns — weighting the frame by R1-6 sensitivity therefore *lowers* retinal signal relative to human luma. *Palette*: because R1-6 are a single spectral class the motion pathway is monochromatic, so hue is not a quantity it can use, and we re-emit the scene’s achromatic structure in blue-green — equivalent to repainting every surface, and the same move a fly-vision laboratory makes when it builds an arena from green emitters. That recovers the lost signal and then some, raising retinal standard deviation from 0.020 to 0.033, measured open loop on 100 identical frames.

None of it releases LPLC2. We note the engine’s own display controls are not a route either: `vid_gamma`, `vid_contrast`, `vid_brightness` and `vid_saturation` leave the rendered buffer bit-identical, because ViZDoom exposes it upstream of display post-processing, and `r_visibility`, which does take effect, moves frame mean brightness without changing the distance-to-brightness relation by more than 3–6% over a 40-tic approach. Every one of these is a common scale on both arms of the imbalance, which is why the ratio argument predicts the outcome in each case.

Rate readout. An exponential filter over per-step output. Because that output is $\text{rate} \times dt$, recovering Hz requires division by dt ; dividing by τ instead scales every reported rate by $160\times$ and is a failure mode we hit and document. A 12-tic warmup precedes baseline estimation, and a $\tau = 3$ s baseline removes the standing left-right asymmetry.

Statistics. The olfaction comparison uses 10 independent environment seeds. Each seed contributes one paired on/off difference, averaged over 320 tics after a 40-tic settling period that lets the rate filter converge. We report the mean and standard deviation across seeds together with a two-sided

sign test on the number agreeing with the mean direction ($p = 2 \sum_{i=k}^n \binom{n}{i} / 2^n$; $p = 0.002$ at 10/10, $p = 0.021$ at 9/10). We prefer the sign test to a t -test because per-seed effect magnitude depends on how closely enemies happen to approach, which is neither normal nor identically distributed across seeds — it is a property of the episode, not of the model. Note that sign consistency alone is a weak claim when an effect is small, which is why we read the LC4 control on magnitude rather than on its p -value. The degree-preserving shuffle permutes postsynaptic targets while preserving each neuron’s in-degree, and is run through the identical protocol with the identical seeds. Table and figure are generated from the same serialized run data, so they cannot diverge.

The environment is not bit-reproducible across processes, so the magnitudes carry run-to-run variation of a few percent and the sign counts are less stable than a single execution suggests. We can put a number on the latter: correcting an error in the odour-source table — `CustomMedikit`, the pickup actor used by one scenario, was absent from the food list, and unlisted actors fell through to the pheromone channel, so inanimate scenery was being smelled as conspecifics — left the lateral horn response essentially unchanged in size but moved `DNp01` and `DNa02` from 10/10 agreeing seeds to 9/10. Sign counts near ceiling should therefore be read as p values with one seed of slack, not as invariants.

Tested, and deferred

Recording what was tried and failed is as much a part of the result as what worked, so we list both.

Tested, and does not rescue visual motion: per-type conduction delays (1.8–240 ms); per-neuron membrane time constants (5–200 ms); conductance-based shunting synapses, which are now the default rather than a candidate; a forced high-conductance state; population pooling; graded non-spiking optic-lobe units; tonic optic-lobe bias from 0 to 7.5 mV; photoreceptor drive, graded rate ceiling and Weber contrast gain; injection at the photoreceptors rather than the lamina; field of view from 110° to 170°; sRGB linearisation; R1-6 spectral weighting; repainting the environment into the fly’s band; and the engine’s own rendering controls. The common structure of these failures is that each is a scale applied to both arms of an imbalance, and a scale cancels from a ratio.

Corrected during this work, all on the input side: the retina mapped azimuth linearly onto a perspective projection, misplacing a column at 10° of gaze by 87%; the field of view was never applied, because ViZDoom ignores it as a command-line argument, so the model ran at the engine default rather than the reported value; frames were held across all 57 substeps of a tic, presenting a correlator with an impulse train aliased against its own 80 ms delay line; and, consequently, Weber adaptation advanced once per tic rather than once per substep, giving it an effective time constant of 14 s in place of 0.25 s. Each of these corrupts precisely the cue under test. None of them changed the outcome, which is itself informative: the negative result does not rest on a badly delivered stimulus.

Deferred. Per-cell-type synaptic gains are the identified obstacle and we do not fit them, because fitting them is the independent variable of this study rather than a repair to it. We note the target is not a single set of numbers: the optimized model of [Lappalainen et al. \[2024\]](#) carries 734 free parameters, 604 of them one gain per type-to-type connection, and its authors report an ensemble of 50 solutions of comparable task performance, so connectome plus task does not identify them uniquely. Also deferred: gap junctions, absent from the reconstruction and carrying the giant fibre’s fastest output; neuromodulation, flattened here to fast transmission; and a ventral nerve cord, without which every behavioural timescale is the engine’s.

Data and code availability

Code is available at <https://github.com/mutkuoz/flydoom> under the MIT licence. The connectome data are CC BY-NC 4.0, are not redistributed, and must be obtained from FlyWire directly; a fetch script is provided. The environment uses ViZDoom [Kempka et al., 2016] with the freely licensed Freedoom asset set, so no commercial game data are required.

DOOM is a trademark of id Software LLC / ZeniMax Media Inc. This work is unaffiliated with and unendorsed by them, and the name is used here only to identify the software environment.

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References

- Salil S. Bidaye, Christian Machacek, Yang Wu, and Barry J. Dickson. Neuronal control of *Drosophila* walking direction. *Science*, 344(6179):97–101, 2014. doi: 10.1126/science.1249964.
- Salil S. Bidaye, Meghan Laturney, Amy K. Chang, Yuejiang Liu, Till Bockemühl, Ansgar Büschges, and Kristin Scott. Two brain pathways initiate distinct forward walking programs in *Drosophila*. *Neuron*, 108(3):469–485.e8, 2020. doi: 10.1016/j.neuron.2020.07.032.
- Alexander Borst, Juergen Haag, and Dierk F. Reiff. Fly motion vision. *Annual Review of Neuroscience*, 33(1):49–70, 2010. doi: 10.1146/annurev-neuro-060909-153155.
- Julia Buhmann, Arlo Sheridan, Caroline Malin-Mayor, Philipp Schlegel, Stephan Gerhard, Tom Kazimiers, Renate Krause, Tri M. Nguyen, et al. Automatic detection of synaptic partners in a whole-brain *Drosophila* electron microscopy data set. *Nature Methods*, 18(7):771–774, 2021. doi: 10.1038/s41592-021-01183-7.
- Thomas A. Cleland. Inhibitory glutamate receptor channels. *Molecular Neurobiology*, 13(2):97–136, 1996. doi: 10.1007/bf02740637.
- Michael-John Dolan, Shahar Frechter, Alexander Shakeel Bates, Chuntao Dan, Paavo Huoviala, Ruairí JV Roberts, Philipp Schlegel, Serene Dhawan, et al. Neurogenetic dissection of the *Drosophila* lateral horn reveals major outputs, diverse behavioural functions, and interactions with the mushroom body. *eLife*, 8:e43079, 2019. doi: 10.7554/elife.43079.
- Sven Dorkenwald, Arie Matsliah, Amy R. Sterling, Philipp Schlegel, Szi-chieh Yu, Claire E. McKellar, Albert Lin, Marta Costa, et al. Neuronal wiring diagram of an adult brain. *Nature*, 634(8032):124–138, 2024. doi: 10.1038/s41586-024-07558-y.
- Nils Eckstein, Alexander Shakeel Bates, Andrew Champion, Michelle Du, Yijie Yin, Philipp Schlegel, Alicia Kun-Yang Lu, Thomson Rymer, et al. Neurotransmitter classification from electron microscopy images at synaptic sites in *Drosophila melanogaster*. *Cell*, 187(10):2574–2594.e23, 2024. doi: 10.1016/j.cell.2024.03.016.
- Eyal Gruntman, Sandro Romani, and Michael B. Reiser. Simple integration of fast excitation and offset, delayed inhibition computes directional selectivity in *Drosophila*. *Nature Neuroscience*, 21(2):250–257, 2018. doi: 10.1038/s41593-017-0046-4.

- Juergen Haag, Alexander Arenz, Etienne Serbe, Fabrizio Gabbiani, and Alexander Borst. Complementary mechanisms create direction selectivity in the fly. *eLife*, 5:e17421, 2016. doi: 10.7554/elife.17421.
- R. C. Hardie. A histamine-activated chloride channel involved in neurotransmission at a photoreceptor synapse. *Nature*, 339(6227):704–706, 1989. doi: 10.1038/339704a0.
- B. Hassenstein and W. Reichardt. Systemtheoretische Analyse der Zeit-, Reihenfolgen- und Vorzeichenbewertung bei der Bewegungsperzeption des Rüsselkäfers *Chlorophanus*. *Zeitschrift für Naturforschung B*, 11(9-10):513–524, 1956. doi: 10.1515/znb-1956-9-1004.
- Gregory S.X.E. Jefferis, Christopher J. Potter, Alexander M. Chan, Elizabeth C. Marin, Torsten Rohlffing, Calvin R. Maurer, and Liqun Luo. Comprehensive maps of *Drosophila* higher olfactory centers: Spatially segregated fruit and pheromone representation. *Cell*, 128(6):1187–1203, 2007. doi: 10.1016/j.cell.2007.01.040.
- Kyobi S. Kakaria and Benjamin L. de Bivort. Ring attractor dynamics emerge from a spiking model of the entire protocerebral bridge. *Frontiers in Behavioral Neuroscience*, 11:8, 2017. doi: 10.3389/fnbeh.2017.00008.
- Michal Kempka, Marek Wydmuch, Grzegorz Runc, Jakub Toczek, and Wojciech Jaskowski. ViZDoom: A Doom-based AI research platform for visual reinforcement learning. In *2016 IEEE Conference on Computational Intelligence and Games (CIG)*, pages 1–8, 2016. doi: 10.1109/cig.2016.7860433.
- Amina Kurtovic, Alexandre Widmer, and Barry J. Dickson. A single class of olfactory neurons mediates behavioural responses to a *Drosophila* sex pheromone. *Nature*, 446(7135):542–546, 2007. doi: 10.1038/nature05672.
- Janne K. Lappalainen, Fabian D. Tschopp, Sridhama Prakhya, Mason McGill, Aljoscha Nern, Kazunori Shinomiya, Shin-ya Takemura, Eyal Gruntman, et al. Connectome-constrained networks predict neural activity across the fly visual system. *Nature*, 634(8036):1132–1140, 2024. doi: 10.1038/s41586-024-07939-3.
- Matthew S. Maisak, Juergen Haag, Georg Ammer, Etienne Serbe, Matthias Meier, Aljoscha Leonhardt, Tabea Schilling, Armin Bahl, et al. A directional tuning map of *Drosophila* elementary motion detectors. *Nature*, 500(7461):212–216, 2013. doi: 10.1038/nature12320.
- Aleksandr Rayshubskiy, Stephen L Holtz, Alexander S Bates, Quinn X Vanderbeck, Laia Seratosa Capdevila, Victoria Rockwell, and Rachel Wilson. Neural circuit mechanisms for steering control in walking *Drosophila*. *eLife*, 13:RP102230, 2025. doi: 10.7554/eLife.102230.
- Philipp Schlegel, Yijie Yin, Alexander S. Bates, Sven Dorkenwald, Katharina Eichler, Paul Brooks, Daniel S. Han, Marina Gkantia, et al. Whole-brain annotation and multi-connectome cell typing of *Drosophila*. *Nature*, 634(8032):139–152, 2024. doi: 10.1038/s41586-024-07686-5.
- Julia L. Semmelhack and Jing W. Wang. Select *Drosophila* glomeruli mediate innate olfactory attraction and aversion. *Nature*, 459(7244):218–223, 2009. doi: 10.1038/nature07983.
- Philip K. Shiu, Gabriella R. Sterne, Nico Spiller, Romain Franconville, Andrea Sandoval, Joie Zhou, Neha Simha, Chan Hyuk Kang, et al. A *Drosophila* computational brain model reveals sensorimotor processing. *Nature*, 634(8032):210–219, 2024. doi: 10.1038/s41586-024-07763-9.

- Marcel Stimberg, Romain Brette, and Dan FM Goodman. Brian 2, an intuitive and efficient neural simulator. *eLife*, 8:e47314, 2019. doi: 10.7554/elife.47314.
- Catherine R von Reyn, Patrick Breads, Martin Y Peek, Grace Zhiyu Zheng, W Ryan Williamson, Alyson L Yee, Anthony Leonardo, and Gwyneth M Card. A spike-timing mechanism for action selection. *Nature Neuroscience*, 17(7):962–970, 2014. doi: 10.1038/nn.3741.
- Sibo Wang-Chen, Victor Alfred Stimpfling, Thomas Ka Chung Lam, Pembe Gizem Özdil, Louise Genoud, Femke Hurtak, and Pavan Ramdya. NeuroMechFly v2: simulating embodied sensorimotor control in adult *Drosophila*. *Nature Methods*, 21(12):2353–2362, 2024. doi: 10.1038/s41592-024-02497-y.